

Capacitor

Question1

If a dielectric slab of dielectric constant 3 is introduced between the plates of a capacitor having electric field $15\pi\text{NC}^{-1}$, then the electric displacement is

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Options:

A.

$$1250 \times 10^{-12} \text{Cm}^{-2}$$

B.

$$250 \times 10^{-12} \text{Cm}^{-2}$$

C.

$$125 \times 10^{-9} \text{Cm}^{-2}$$

D.

$$250 \times 10^{-9} \text{Cm}^{-2}$$

Answer: A

Solution:

Given data

- Dielectric constant $K = 3$
- Electric field $E = 15\pi \text{ N/C}$
- We are to find electric displacement D .

Step 1. Recall the relation

$$D = \varepsilon_0 E + P = \varepsilon_0 K E$$

(using $K = \frac{D}{\varepsilon_0 E}$).

So,

$$D = \varepsilon_0 K E$$

Step 2. Substitute values

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$D = (8.854 \times 10^{-12}) \times 3 \times (15\pi)$$

Step 3. Compute

$$D = 8.854 \times 10^{-12} \times 45\pi$$

$$45\pi \approx 141.37$$

$$D = 8.854 \times 10^{-12} \times 141.37$$

$$D \approx 1.252 \times 10^{-9} \text{ C/m}^2$$

Step 4. Express in given unit format

$$1.252 \times 10^{-9} \text{ C/m}^2 = 1250 \times 10^{-12} \text{ C/m}^2$$

 **Final Answer:**

$D = 1250 \times 10^{-12} \text{ C/m}^2$

Correct Option: A

Question2

A capacitor of capacitance $2\mu\text{ F}$ is charged with the help of a 60 V battery. After disconnecting the battery, if this capacitor is connected in parallel with another uncharged capacitor of capacitance $l\mu\text{ F}$, then the potential difference across the plates of $2\mu\text{ F}$ capacitor is

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Options:

A.

30 V

B.

60 V

C.

40 V

D.

20 V

Answer: C

Solution:

Step 1: Find the charge on the first capacitor.

The $2\ \mu\text{F}$ capacitor is charged using the 60 V battery, so its charge is:

$$q_1 = C \times V = 2 \times 10^{-6} \times 60 = 120 \times 10^{-6}\ \text{C}$$

Step 2: Find the charge on the second capacitor.

The $1\ \mu\text{F}$ capacitor is not charged yet, so $q_2 = 0\ \text{C}$

Step 3: Find total charge after connection.

The total charge when we join them is the sum of both charges: $q_{\text{total}} = 120 \times 10^{-6} + 0 = 120 \times 10^{-6}\ \text{C}$

Step 4: Find total capacitance when connected in parallel.

When capacitors are connected in parallel, their capacitances add:

$$C_{\text{total}} = 2\ \mu\text{F} + 1\ \mu\text{F} = 3\ \mu\text{F} = 3 \times 10^{-6}\ \text{F}$$

Step 5: Find the new (final) voltage across the capacitors.

The new voltage is the total charge divided by the total capacitance: $V_{\text{final}} = \frac{q_{\text{total}}}{C_{\text{total}}} = \frac{120 \times 10^{-6}}{3 \times 10^{-6}} = 40\ \text{V}$

Question3

The energy stored in a capacitor of capacitance $10\ \mu\text{F}$ when charged to a potential of 6 kV is

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Options:

A.

100 J

B.

200 J

C.

180 J

D.

160 J

Answer: C

Solution:

Energy stored in a capacitor is,

$$U = \frac{1}{2}CV^2$$

$$\Rightarrow U = \frac{1}{2} \times 10 \times 10^{-6} \times (6 \times 10^3)^2$$

$$= \frac{1}{2} \times 10 \times 10^{-6} \times 36 \times 10^6$$

$$= 180 \text{ J}$$

Question4

A parallel plate capacitor has plates of area $0.4\pi \text{ m}^2$ and spacing of 0.5 mm . If a slab of thickness 0.5 mm and dielectric constant 4.5 is introduced in between the plates of the capacitor, then the capacitance of the capacitor is

Options:

A.

100 nF

B.

60 pF

C.

100 pF

D.

60 nF

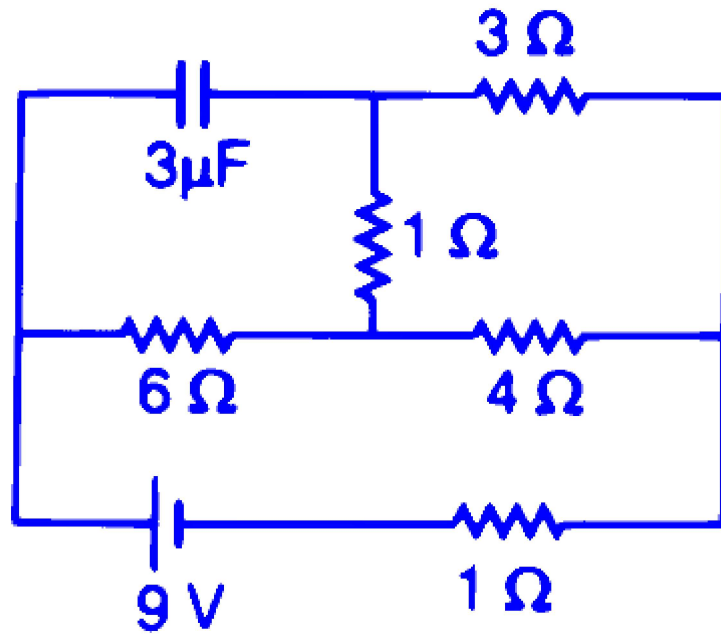
Answer: A**Solution:**

Here dielectric fills the space between plates completely so,

$$\begin{aligned} C &= \frac{\epsilon_0 K A}{d} \\ &= \frac{8.85 \times 10^{-12} \times 4.5 \times 0.4\pi}{0.5 \times 10^{-3}} \\ &\approx 100\text{nF} \end{aligned}$$

Question5

In the given circuit, the potential difference across the plates of the capacitor C in steady state is



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Options:

A.

6.5 V

B.

6 V

C.

9 V

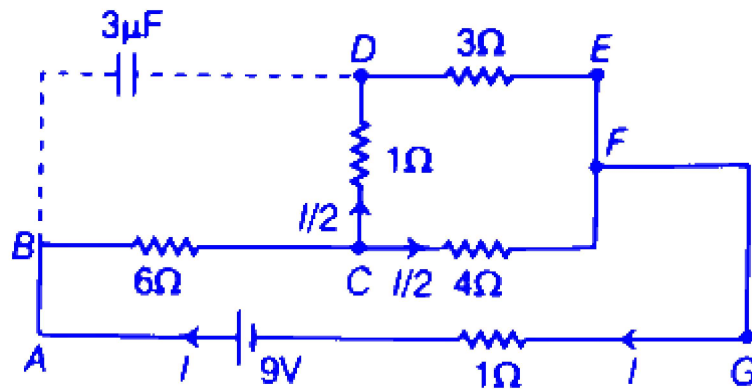
D.

7.5 V

Answer: A

Solution:

In given circuit capacitor does not conduct once it is charged. So in steady state we have following circuit.



$\Rightarrow$ Now, let I = current drawn from cell. Then we have above current distribution.

Using KVL in loop $ABCDEFGA$; we have;

$$-6I - 1 \times \frac{I}{2} - \frac{3I}{2} - I + 9 = 0$$

$$\Rightarrow 9I = 9 \text{ or } I = 1 \text{ A}$$

Now, potential difference between B and D (using KVL);

$$V_B - 6 \times 1 - 1 \times \frac{1}{2} = V_D$$

$$\Rightarrow V_B - V_D = 6 + \frac{1}{2} = \frac{13}{2} = 6.5 \text{ V.}$$

Question6

One of the two identical capacitors having the same capacitance C , is charged to a potential V_1 and the other is charged to a potential V_2 . If they are connected with their like plates together, then the decrease in the electrostatic potential energy of the combined system is

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Options:

A.

$$\frac{C}{4}(V_1^2 - V_2^2)$$

B.

$$\frac{C}{4}(V_1^2 + V_2^2)$$

C.

$$\frac{C}{4}(V_1 - V_2)^2$$

D.

$$\frac{C}{4}(V_1 + V_2)^2$$

Answer: C

Solution:

$$\begin{aligned} U_i &= \frac{1}{2}CV_1^2 + \frac{1}{2}CV_2^2 \\ &= \frac{1}{2}C(V_1^2 + V_2^2) \end{aligned}$$

Common potential,

$$\begin{aligned} V &= \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{CV_1 + CV_2}{C + C} \\ &= \frac{V_1 + V_2}{2} \\ \Rightarrow V &= \frac{V_1 + V_2}{2} \\ \therefore U_f &= \frac{1}{2}C^1V^2 \\ &= \frac{1}{2}(2C)\left(\frac{V_1 + V_2}{2}\right)^2 = C\left(\frac{V_1 + V_2}{2}\right)^2 \\ \therefore \Delta U &= U_i - U_f \\ &= \frac{1}{2}C(V_1^2 + V_2^2) - C\left(\frac{V_1 + V_2}{2}\right)^2 \\ &= \frac{C}{2}(V_1^2 + V_2^2) - \frac{C}{4}(V_1^2 + V_2^2 + 2V_1V_2) \\ &= \frac{C}{4}[2V_1^2 + 2V_2^2 - V_1^2 - V_2^2 - 2V_1V_2] \\ &= \frac{C}{4}[V_1^2 + V_2^2 - 2V_1V_2] = \frac{C}{4}(V_1 - V_2)^2 \end{aligned}$$

Question7

If 27 identical charged conducting spheres each of capacitance $10\mu\text{ F}$ combine to form a big sphere, then the capacitance of the big sphere is

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Options:

A.

$$30\mu \text{ F}$$

B.

$$270\mu \text{ F}$$

C.

$$90\mu \text{ F}$$

D.

$$10\mu \text{ F}$$

Answer: A

Solution:

Capacitance of small conducting sphere

$$C = 4\pi\epsilon_0 r = 10\mu \text{ F}$$

$$\text{Since, } 27 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow R = 3r$$

$\therefore$ Capacitance of big sphere

$$\begin{aligned} C' &= 4\pi\epsilon_0 R = 4\pi\epsilon_0 (3r) \\ &= 3(4\pi\epsilon_0 r) = 3 \times 10 = 30\mu \text{ F} \end{aligned}$$

Question8

The capacitance of a spherical capacitor is 100 pF . If the spacing between the two spheres is 1 cm , then the radius of the inner sphere of the capacitor is

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Options:

A.

9 cm

B.

10 cm

C.

19 cm

D.

20 cm

Answer: A

Solution:

$$\begin{aligned}b - a &= 1 \text{ cm} = 1 \times 10^{-2} \text{ m}, \\b &= (a + 1 \times 10^{-2}) \text{ m} \\C &= 4\pi\epsilon_0 \left(\frac{ab}{b - a} \right) \\\Rightarrow 100 \times 10^{-12} &= \frac{1}{9 \times 10^9} \times \frac{a(a + 1 \times 10^{-2})}{1 \times 10^{-2}} \\9 \times 10^{-3} &= a^2 + 10^{-2}a \\\Rightarrow 0.9 &= 100a^2 + a \\\Rightarrow 100a^2 + a - 0.9 &= 0 \\\Rightarrow a &= \frac{-1 \pm \sqrt{1^2 - 4 \times 100(-0.9)}}{2 \times 100} \\&= -1 \pm \frac{\sqrt{361}}{200} = \frac{-1 \pm 19}{200} \\&= \frac{-1 + 19}{200} = \frac{18}{200} \text{ m} \\&= 0.09 \text{ m} = 9 \text{ cm}\end{aligned}$$

Question9

The energy stored in a capacitor is W . To double the charge on the plates of the capacitor, the additional work to be done is

Options:

A.

W

B.

$4W$

C.

$\frac{4}{3}W$

D.

$3W$

Answer: D

Solution:

$$W = \frac{Q^2}{2C}$$

$$W' = \frac{(2Q)^2}{2C} = 4 \left(\frac{Q^2}{2C} \right) = 4W$$

∴ Additional work done,

$$W' - W = 4W - W = 3W$$

Question10

A wire of length 10 m carrying current of 1 A is bent in to a circular loop. If a magnetic field of $2\pi \times 10^{-4}$ T is applied on the loop, then the maximum torque acting on it is

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Options:

A.

$100 \times 10^{-4} \text{ N} - \text{m}$

B.

$$50 \times 10^{-4} \text{ N} - \text{m}$$

C.

$$25 \times 10^{-4} \text{ N} - \text{m}$$

D.

$$75 \times 10^{-4} \text{ N} - \text{M}$$

Answer: B

Solution:

$$\tau_{\max} = MB$$

$$\tau_{\max} = IAB$$

Since, $2\pi r = 10$

$$\Rightarrow r = \frac{10}{2\pi}$$

$\therefore$ Area of the loop

$$\begin{aligned} A &= \pi r^2 = \pi \left(\frac{10}{2\pi} \right)^2 \\ &= \frac{100}{4\pi} = \frac{25}{\pi} \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \therefore \tau_{\max} &= LAB = 1 \times \frac{25}{\pi} \times 2\pi \times 10^{-4} \\ &= 50 \times 10^{-4} \text{ N} - \text{m} \end{aligned}$$

Question11

The radii of the inner and outer spheres of a spherical capacitor are 8 cm and 9 cm respectively. The outer sphere is earthed and the inner sphere is charged. If the space between the concentric spheres is filled with a liquid of dielectric constant 5 , the capacitance of the capacitor is

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Options:

A.

400 PF

B.

40 PF

C.

$400\mu\text{ F}$

D.

$40\mu\text{ F}$

Answer: A

Solution:

$$\begin{aligned}C &= 4\pi\epsilon_0 k \frac{r_1 r_2}{r_2 - r_1} \\&= 4\pi (8.85 \times 10^{-12}) \times 5 \times \frac{(0.08)(0.09)}{0.09 - 0.08} \\&\simeq 400 \times 10^{-12} \text{ F} \\&\simeq 400\text{pF}\end{aligned}$$

Question12

A capacitor of capacitance $2\mu\text{ F}$ is charged to 50 V and then disconnected from the source. Later the gap between the plates of the capacitor is filled with a dielectric material. If the energy stored in the capacitor is decreased by 25% of its initial value, then the dielectric constant of the dielectric material is

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Options:

A.

B.

$$\frac{4}{3}$$

C.

$$\frac{3}{4}$$

D.

$$\frac{3}{2}$$

Answer: B

Solution:

Initial stored energy

$$E_i = \frac{1}{2}CV^2$$

Final stored energy,

$$E_f = E_i - 0.25E_i$$

$$E_f = 0.75E_i \quad \dots (i)$$

$$\therefore E_f = \frac{1}{2}C'V'^2$$

$$= \frac{1}{2}(KC)\left(\frac{V}{k}\right)^2$$

$$E_f = \frac{E_i}{K}$$

$$\Rightarrow 0.75E_i = \frac{E_i}{K} \quad [\text{from eq. (i)}]$$

$$\Rightarrow \frac{1}{K} = 0.75$$

$$\Rightarrow K = \frac{1}{0.75} = \frac{4}{3}$$

Question13

The space between the plates of a parallel plate capacitor is halved and a dielectric medium of relative permittivity 10 is introduced between the plates. The ratio of the final and initial capacitances of the capacitor is

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Options:

- A. 20
- B. 10
- C. 1/10
- D. 1/20

Answer: A

Solution:

The capacitance of a parallel plate capacitor is given by the formula:

$$C = \frac{A\epsilon_0}{d}$$

where:

A is the area of the plates,

ϵ_0 is the permittivity of free space,

d is the distance between the plates.

In this scenario, the distance between the plates is halved, and a dielectric material with a relative permittivity (k) of 10 is introduced. The new capacitance (C') can be calculated as follows:

$$C' = \frac{kA\epsilon_0}{d'} = \frac{10 \cdot A\epsilon_0}{d/2}$$

Simplifying gives:

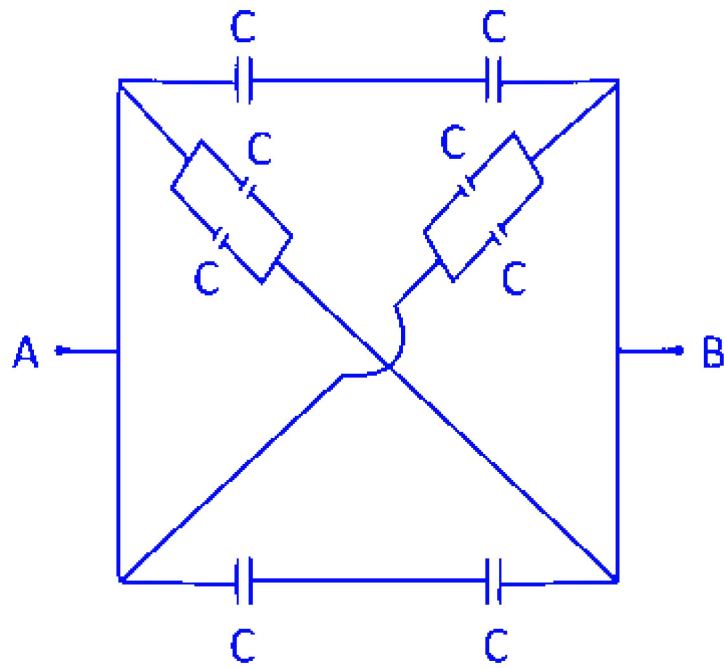
$$C' = \frac{10 \cdot A\epsilon_0}{d/2} = \frac{20 \cdot A\epsilon_0}{d} = 20C$$

Thus, the ratio of the final capacitance to the initial capacitance is:

$$\frac{C'}{C} = 20$$

Question14

Eight capacitors each of capacity 2μ F are arranged as shown in figure. The effective capacitance between A and B is



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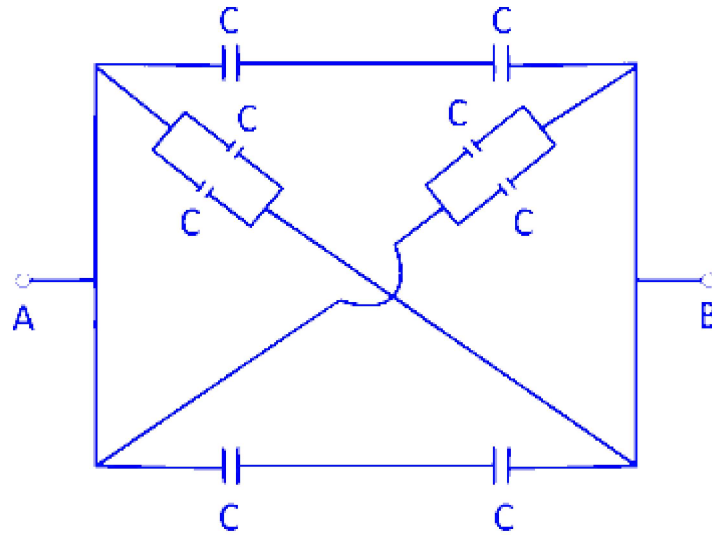
Options:

- A. $10\mu\text{ F}$
- B. $12\mu\text{ F}$
- C. $16\mu\text{ F}$
- D. $4\mu\text{ F}$

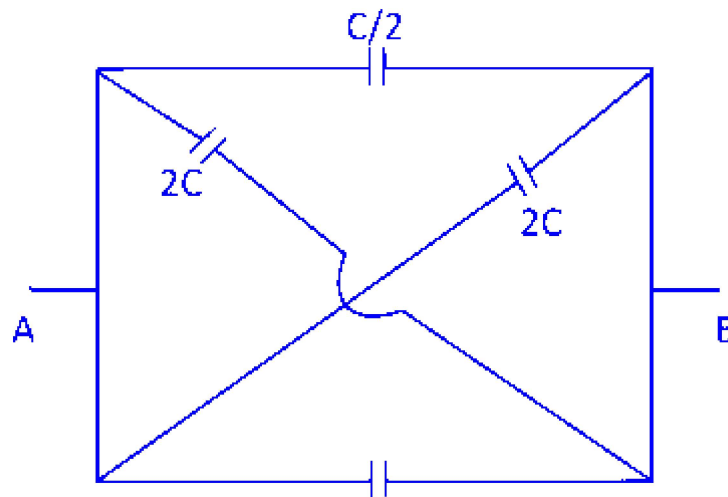
Answer: A

Solution:

Given capacitance, $C = 2\mu\text{ F}$



Equivalent circuit, Thus, $C_{eq} = \frac{C}{2} + \frac{C}{2} + 2C + 2C$



$$= C + 4C$$

$$C_{eq} = 5C$$

$$\therefore C_{eq} = 5 \times 2 = 10\mu F$$

Question15

When a parallel plate capacitor is charged up to 95 V , its capacitance is C . If a dielectric slab of thickness 2 mm is inserted between plates and distance between the plates is increased by 1.6 mm such that the same potential difference is maintained. The dielectric constant of the material (slab) is

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Options:

A. 2.4

B. 4.5

C. 5.0

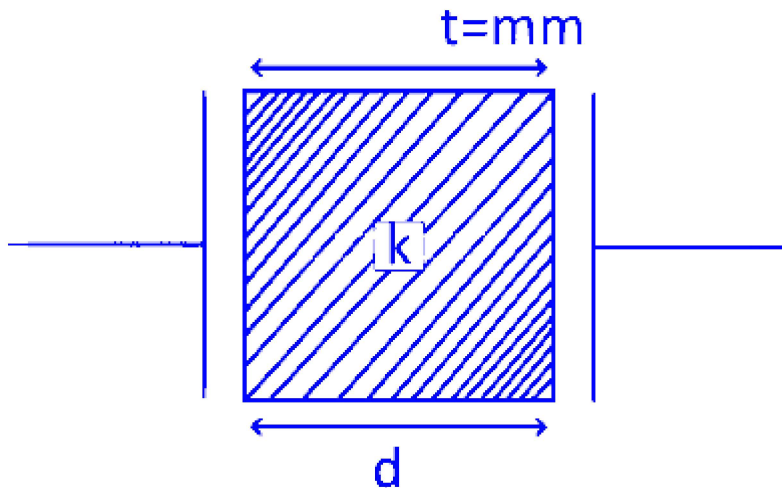
D. 9.0

Answer: C

Solution:

Let initial thickness = d

Thickness of dielectrics = 2 mm



It is given that final potential is same.

Let initial capacitance = C_1

Final capacitance = C_2

$$\therefore C_1 = C_2 \quad \dots (i)$$

$$C_1 = \frac{A\epsilon_0}{d} \text{ and } C_2 = \frac{A\epsilon_0}{d' - t + \frac{t}{k}}$$

where, $d' = d + 1.6 \text{ mm}$

$$\Rightarrow \frac{A\epsilon_0}{d} = \frac{A\epsilon_0}{(d + 1.6) - t \left(1 - \frac{1}{k}\right)}$$

$$\Rightarrow \frac{1}{d} = \frac{1}{(d + 1.6) - 2 \left(1 - \frac{1}{k}\right)}$$

$$\Rightarrow d + 1.6 - 2 + \frac{2}{k} = d$$

$$\Rightarrow k = 5$$

Question16

The capacitance of an isolated sphere of radius r_1 is increased by 5 times, when it is enclosed by an earthed concentric sphere of radius r_2 . The ratio of their radii is

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Options:

A. $\frac{4}{5}$

B. $\frac{5}{4}$

C. $\frac{5}{1}$

D. $\frac{3}{5}$

Answer: A

Solution:

Given,

Radius of isolated sphere = r_1

Radius of earthed sphere = r_2

Capacitance of isolated sphere,

$$C_1 = 4\pi\epsilon_0 r_1 \quad \dots (i)$$

Capacitance of isolated sphere enclosed in earthed sphere,

$$C_2 = \frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1} \quad \dots (ii)$$

Now, capacitance is 5 times

$$\Rightarrow 5C_1 = C_2$$

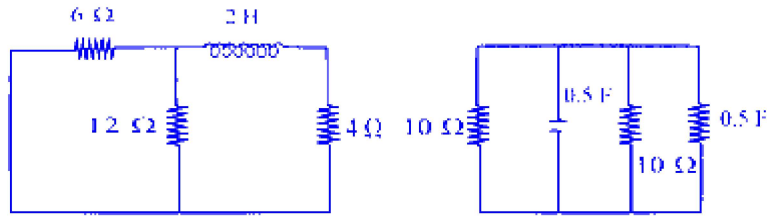
$$\Rightarrow 5(4\pi\epsilon_0 r_1) = \frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1} \Rightarrow 5 = \frac{r_2}{r_2 - r_1}$$

$$\Rightarrow 5r_2 - 5r_1 = r_2 \Rightarrow -5r_1 = -4r_2$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{4}{5}$$

Question17

Two figures are shown as Fig. *A* and Fig. *B*. The time constant of Fig. *A* is τ_A and time constant of Fig. *B* is τ_B . Then



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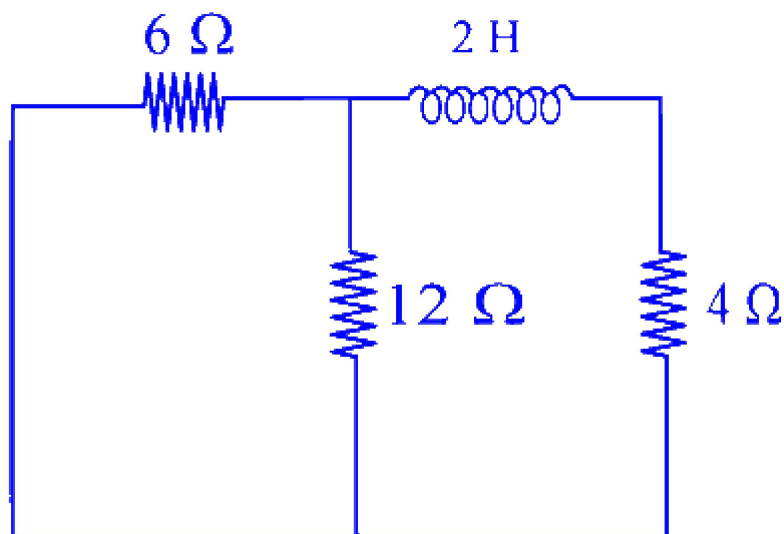
Options:

- A. $\tau_A = \frac{1}{4}\ \text{s}$ and $\tau_B = 5\ \text{s}$
- B. $\tau_A = \frac{1}{2}\ \text{s}$ and $\tau_B = \frac{1}{5}\ \text{s}$
- C. $\tau_A = 4\ \text{s}$ and $\tau_B = 5\ \text{s}$
- D. $\tau_A = \frac{1}{3}\ \text{s}$ and $\tau_B = 10\ \text{s}$

Answer: A

Solution:

For fig (A),

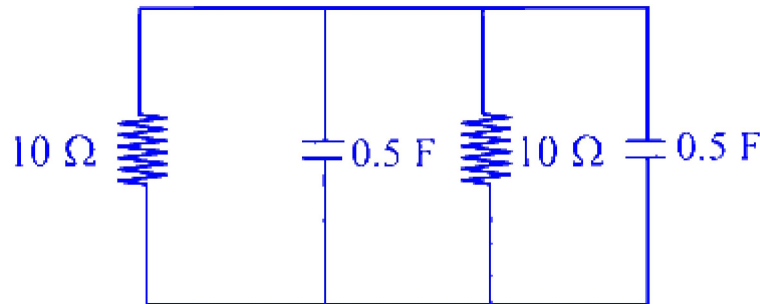


$$R' = \frac{6 \times 12}{6 + 12} + 4 = 4 + 4$$

$$R' = 8$$

$\therefore$ Time constant for circuit (A)

$$\tau_A = \frac{L}{R'} = \frac{2}{8} = \frac{1}{4} \text{ s}$$



Similar for figure (B)

$$R'' = \frac{10 \times 10}{10 + 10} = 5 \Omega$$

and equivalent capacitance,

$$C_{eq} = 0.5 + 0.5 = 1 \text{ F}$$

$\therefore$ Time constant, $\tau_B = RC$

$$= 5 \times 1 = 5 \text{ s}$$

Question18

Three parallel plate capacitors of capacitances $4\mu\text{ F}$, $6\mu\text{ F}$ and $12\mu\text{ F}$ are first connected in series and then in parallel. The ratio of the effective capacitances in the two cases is

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Options:

A. 1 : 11

B. 5 : 8

C. 3 : 7

D. 4 : 9

Answer: A

Solution:

Given the capacitances:

$$C_1 = 4 \mu\text{F}$$

$$C_2 = 6 \mu\text{F}$$

$$C_3 = 12 \mu\text{F}$$

When connected in series:

To find the equivalent capacitance for capacitors in series, we use the formula:

$$C_{\text{eq}} = \frac{C_1 C_2 C_3}{C_1 C_2 + C_2 C_3 + C_3 C_1}$$

Substituting the given values:

$$C_{\text{eq}} = \frac{4 \times 6 \times 12}{4 \times 6 + 6 \times 12 + 12 \times 4} = \frac{288}{144} = 2 \mu\text{F}$$

When connected in parallel:

For capacitors in parallel, the equivalent capacitance is the sum of the capacitances:

$$C_{\text{eq}} = C_1 + C_2 + C_3$$

Substituting the given values:

$$C_{\text{eq}} = 4 + 6 + 12 = 22 \mu\text{F}$$

Calculating the ratio:

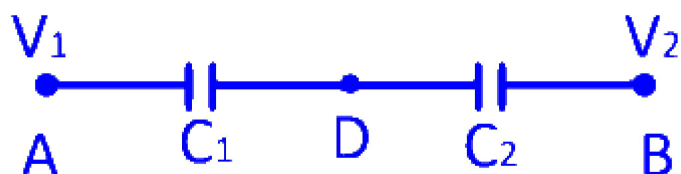
The ratio of the effective capacitances when connected in series to when connected in parallel is:

$$\frac{2}{22} = \frac{1}{11}$$

Thus, the ratio is 1 : 11.

Question 19

Two condensers C_1 and C_2 in a circuit are joined as shown in the figure. The potential of point A is V_1 and that of point B is V_2 . The potential at point D will be



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Options:

A. $\frac{1}{2}(V_1 + V_2)$

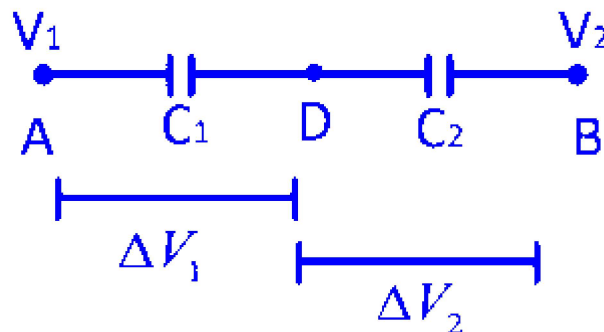
B. $\frac{C_2 V_2 + C_1 V_1}{C_1 + C_2}$

C. $\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

D. $\frac{C_2 V_1 - C_1 V_2}{C_1 + C_2}$

Answer: C

Solution:



Let the potential of point D be V .

In the above figure, capacitors C_1 and C_2 are in series combination.

Potential difference across both the capacitors are,

$$\Delta V_1 = V_1 - V$$

$$\Delta V_2 = V - V_2$$

We know that in series arrangement, the charge on capacitor will be equal.

$$\Rightarrow Q_1 = Q_2$$

$$C_1 (V_1 - V) = C_2 (V - V_2)$$

$$\Rightarrow C_1 V_1 - C_1 V = C_2 V - C_2 V_2$$

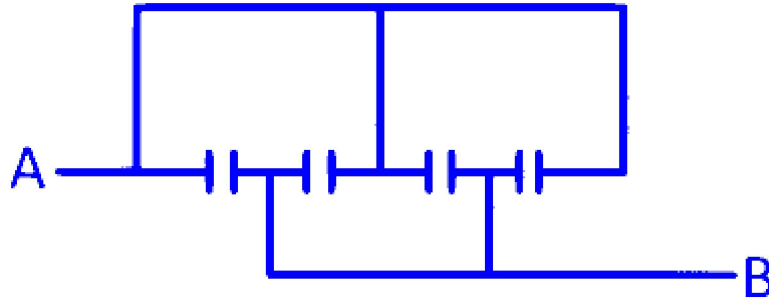
$$\Rightarrow C_1 V + C_2 V = C_2 V_2 + C_1 V_1$$

$$\Rightarrow V (C_1 + C_2) = C_1 V_1 + C_2 V_2$$

$$\Rightarrow V = \frac{C_1 V_1 + C_2 V_2}{(C_1 + C_2)}$$

Question20

Four condensers each of capacitance $8\mu\text{ F}$ are joined as shown in the figure. The equivalent capacitance between the points A and B will be



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Options:

A. $32\mu\text{ F}$

B. $2\mu\text{ F}$

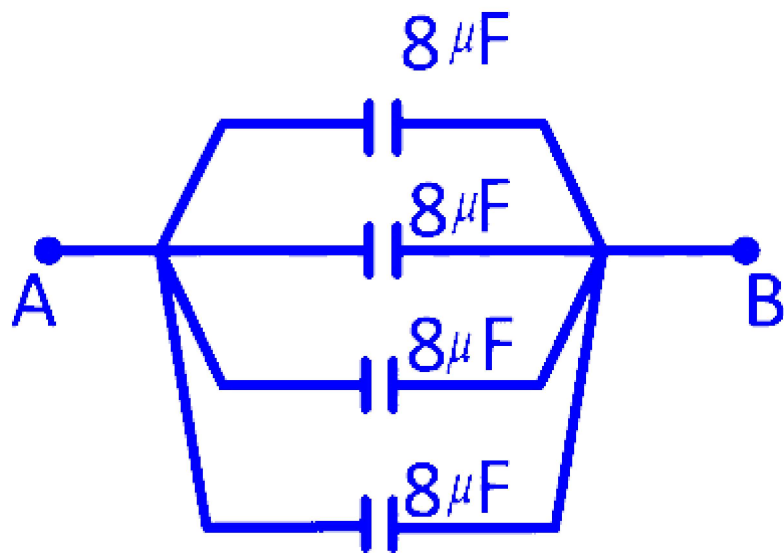
C. $8\mu\text{ F}$

D. $16\mu\text{ F}$

Answer: A

Solution:

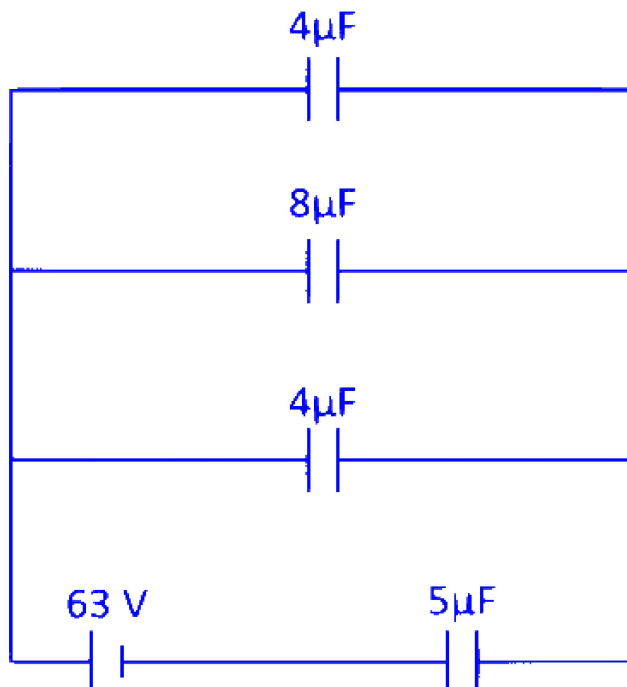
The given arrangement of capacitor are represented as



$$\therefore C_{AB} = 8 + 8 + 8 + 8 = 32\mu\text{F}$$

Question21

In the given circuit, the potential difference across $5\mu\text{F}$ capacitor is



Options:

A. 48 V

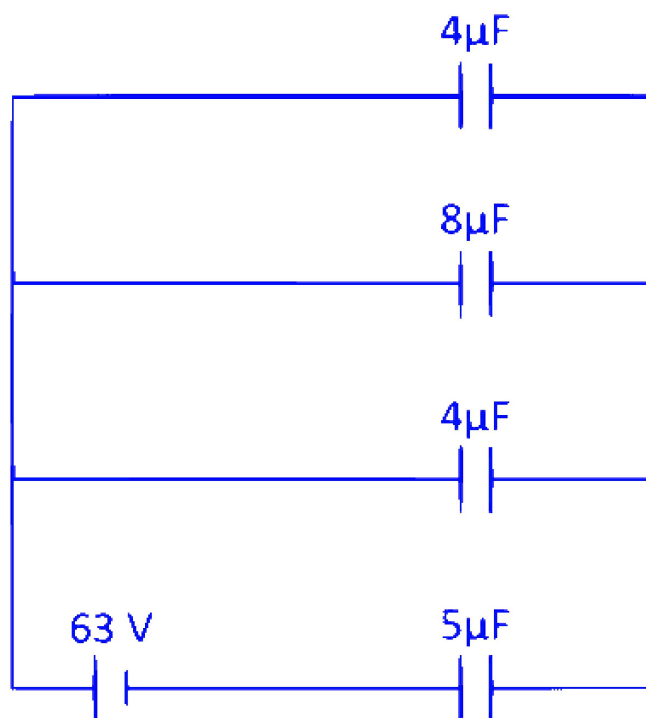
B. 24 V

C. 63 V

D. 21 V

Answer: A

Solution:



According to figure,

$4\mu\text{ F}$, $8\mu\text{ F}$, $4\mu\text{ F}$ are in parallel combination. So,

$$C' = 4 + 8 + 4 = 16\mu\text{ F} \quad \dots \text{ (i)}$$

Now, C' and $5\mu\text{ F}$ are in series combination. So, total capacitance in the circuit is,

$$\frac{1}{C_{\text{total}}} = \frac{1}{5} + \frac{1}{C'}$$

$$\text{From Eq. (i), } C_{\text{total}} = \frac{16 \times 5}{16 + 5} = \frac{80}{21} \mu\text{ F}$$

Total charge in the given circuit,

$$\begin{aligned}
 q_{\text{total}} &= C_{\text{total}} \cdot V \\
 &= \frac{80}{21} \times 63 = 240\mu\text{C}
 \end{aligned}$$

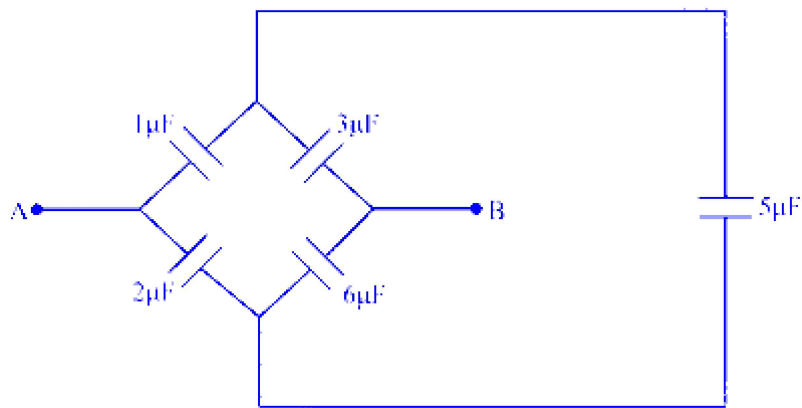
Charge remains same in series combination. So, $5\mu\text{F}$ also have $240\mu\text{C}$ charge.

The potential difference across $5\mu\text{F}$ capacitor is,

$$\begin{aligned}
 V' &= \frac{Q}{C} = \frac{240}{5} \\
 V' &= 48\text{ V}
 \end{aligned}$$

Question22

The capacitance between the points A and B in the following figure.



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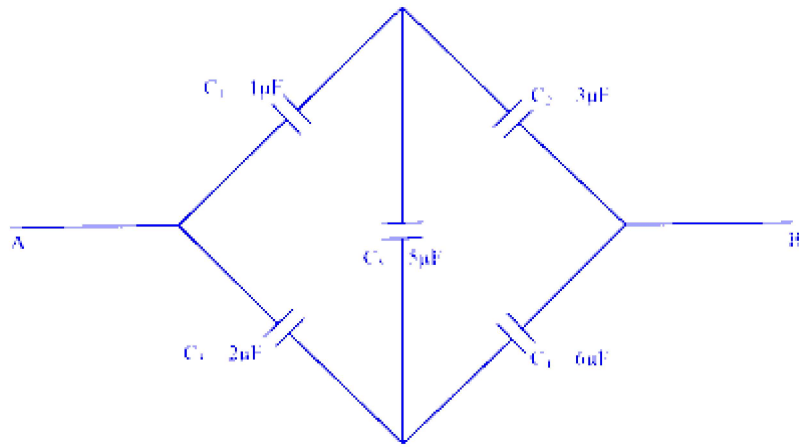
Options:

- A. $\frac{3}{8}\mu\text{F}$
- B. $\frac{9}{4}\mu\text{F}$
- C. $\frac{4}{5}\mu\text{F}$
- D. $2\mu\text{F}$

Answer: B

Solution:

The given circuit diagram can be represented as

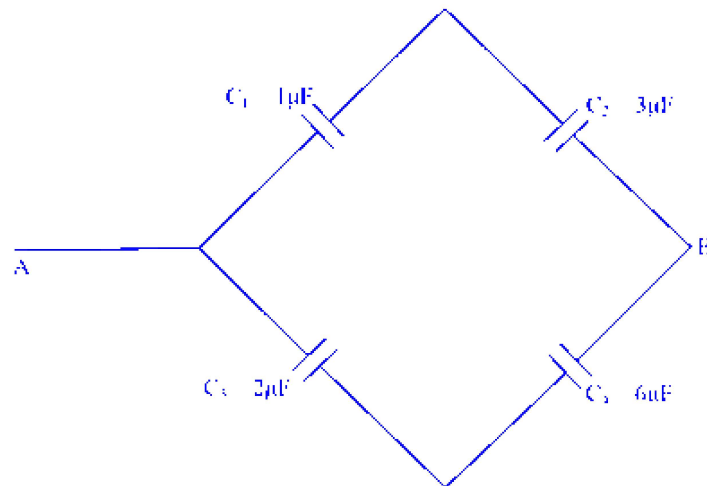


Since, $\frac{C_1}{C_2} = \frac{C_3}{C_4} = \frac{1}{3}$

Hence, given circuit follows the condition of balance wheatstone bridge.

Hence, capacitor C_5 is ineffective.

Now, the circuit diagram is redrawn as

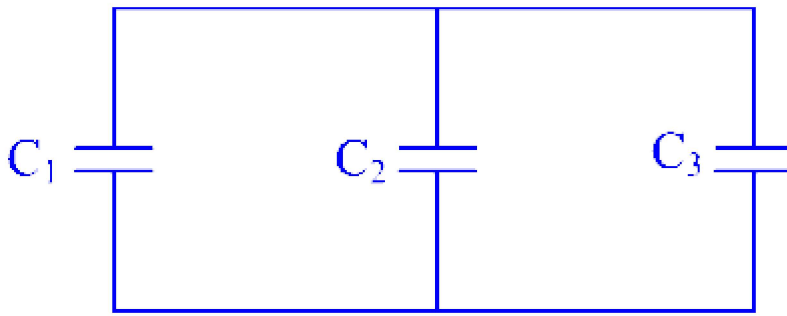


$$C_{AB} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4} = \frac{1 \times 3}{1 + 3} + \frac{2 \times 6}{2 + 6}$$

$$= \frac{3}{4} + \frac{12}{8} = \frac{9}{4} \mu\text{F}$$

Question23

A capacitor of capacitance $C_1 = 1\mu\text{F}$ is charged using a 9 V battery. C_1 is, then removed from the battery and connected to capacitors C_2 and C_3 of $2\mu\text{F}$ and $3\mu\text{F}$, respectively as shown in the figure. Find the charge on C_3 after equilibrium has reached is



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Options:

A. $4.5 \times 10^{-6} \text{C}$

B. $3.5 \times 10^{-6} \text{C}$

C. $2.5 \times 10^{-6} \text{C}$

D. $1.5 \times 10^{-5} \text{C}$

Answer: A

Solution:

Given, $C_1 = 1\mu\text{F}$, $V = 9 \text{ V}$,

$$C_2 = 2\mu\text{F}, C_3 = 3\mu\text{F}$$

Equivalent capacitance of C_2 and C_3

$$C' = C_2 + C_3 = 2 + 3 = 5\mu\text{F}$$

Common potential difference when C_1 and C' are combined, is given as

$$V' = \frac{\text{total charge } (Q)}{\text{total capacitance } (C_1 + C')}$$

$$V' = \frac{Q}{C_1 + C'}$$

$$\text{But } Q = C_1 V$$

$$\therefore V' = \frac{C_1 V}{C_1 + C'} = \frac{1 \times 9}{1 + 5} = \frac{9}{6} = 1.5 \text{ V}$$

Charge on capacitor C_3 is given as

$$\begin{aligned} Q_3 &= C_3 V' \\ &= 3 \times 10^{-6} \times 1.5 = 4.5 \times 10^{-6} \text{C} \end{aligned}$$

Question24

A $60\ \mu\text{F}$ parallel plate capacitor whose plates are separated by $6\ \text{mm}$ is charged to $250\ \text{V}$, and then the charging source is removed. When a slab of dielectric constant 5 and thickness $3\ \text{mm}$ is placed between the plates, find the change in the potential difference across the capacitor.

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Options:

A. $250\ \text{V}$

B. $100\ \text{V}$

C. $150\ \text{V}$

D. $75\ \text{V}$

Answer: B

Solution:

Given,

Capacitance of parallel plate capacitor, $C_0 = 60\ \mu\text{F}$

$$= 60 \times 10^{-6}\ \text{F}$$

Separation between plates, $d = 6\ \text{mm}$

$$= 6 \times 10^{-3}\ \text{m}$$

Electric potential, $V_0 = 250\ \text{V}$

Dielectric constant of slab, $K = 5$

and thickness of slab, $t = 3\ \text{mm}$

$$= 3 \times 10^{-3}\ \text{m}$$

$$\text{Since, } C = \frac{q}{V} = \frac{\epsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}$$

where, q is charge stored, V is potential, ϵ_0 is free space permittivity $= 8.854 \times 10^{-12} \text{Nm}^2\text{C}^{-2}$ and A is plate area

$$\begin{aligned}\therefore q_0 &= C_0 V_0 \\ &= 60 \times 10^{-6} \times 250 = 15 \times 10^{-3} \text{C}\end{aligned}$$

and new capacitance, $C' = \frac{\epsilon_0 A}{[6-3(1-\frac{1}{5})] \times 10^{-3}} \dots (i)$

and $\epsilon_0 A = C_0 d$

$$\begin{aligned}\Rightarrow \epsilon_0 A &= 60 \times 10^{-6} \times 6 \times 10^{-3} \\ &= 360 \times 10^{-9}\end{aligned}$$

Put this value in Eq. (i), we get

$$C' = \frac{360 \times 10^{-9}}{3.6 \times 10^{-3}} = 10^{-4} \text{ F}$$

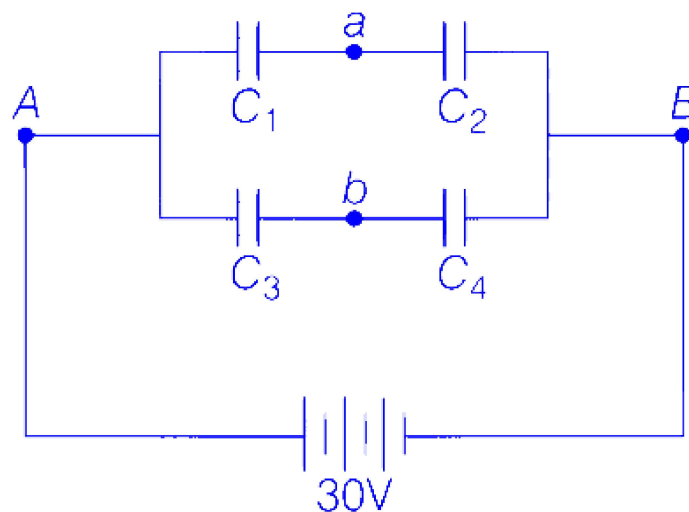
Final potential stored in capacitor $= \frac{q_0}{C'} = \frac{15 \times 10^{-3}}{10^{-4}} = 150 \text{ V}$

Hence, voltage difference $= 250 - 150 = 100 \text{ V}$

Question 25

Four capacitors with capacitances

$C_1 = 1 \text{ } \mu\text{F}$, $C_2 = 1.5 \text{ } \mu\text{F}$, $C_3 = 2.5 \text{ } \mu\text{F}$ and $C_4 = 0.5 \text{ } \mu\text{F}$ are connected as shown and are connected to a 30 V source. The potential difference between points a and b is



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Options:

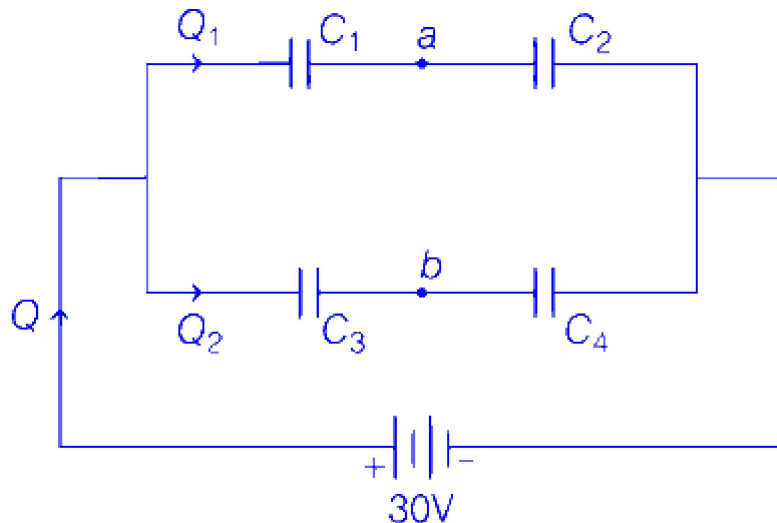
- A. 5 V
- B. 9 V
- C. 10 V
- D. 13 V

Answer: D

Solution:

Given, C_1, C_2, C_3 and C_4 are $1\ \mu\text{F}$, $1.5\ \mu\text{F}$, $2.5\ \mu\text{F}$ and $0.5\ \mu\text{F}$.

Supply voltage = 30 V



Let V_{ab} is potential across a and b ,

Q is charge in circuit and Q_1, Q_2 are branch charges.

Since, series equivalent capacitance $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$

$$\Rightarrow C_s = \frac{C_1 C_2}{C_1 + C_2}$$

and in parallel connection voltages remain same

$$\therefore Q_1 = \frac{C_1 C_2}{C_1 + C_2} V \Rightarrow Q_1 = \frac{1 \times 1.5}{1 + 1.5} \times 30$$

$$= \frac{1.5}{2.5} \times 30 = 18\ \mu\text{C}$$

$$\text{Similarly, } Q_2 = \frac{2.5 \times 0.5}{2.5 + 0.5} \times 30 = 12.5\ \mu\text{C}$$

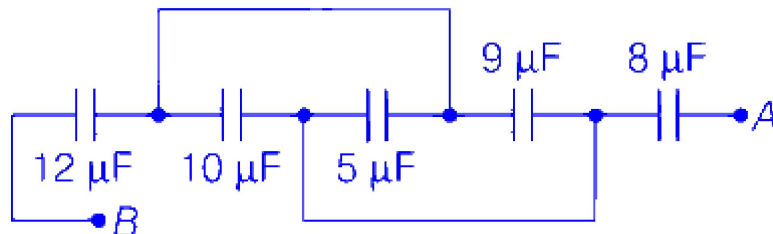
$$\therefore V_{ab} = V_a - V_b = (V - V_b) - (V - V_a)$$

$$= \frac{Q_2}{C_3} - \frac{Q_1}{C_1} = \frac{12.5}{2.5} - \frac{18}{1} = -13 \text{ V}$$

$$\therefore V_{ba} = 13 \text{ V}$$

Question 26

In the given circuit, if the potential difference between A and B is 80 V, then the equivalent capacitance between A and B and the charge on $10 \mu\text{F}$ capacitor respectively, are



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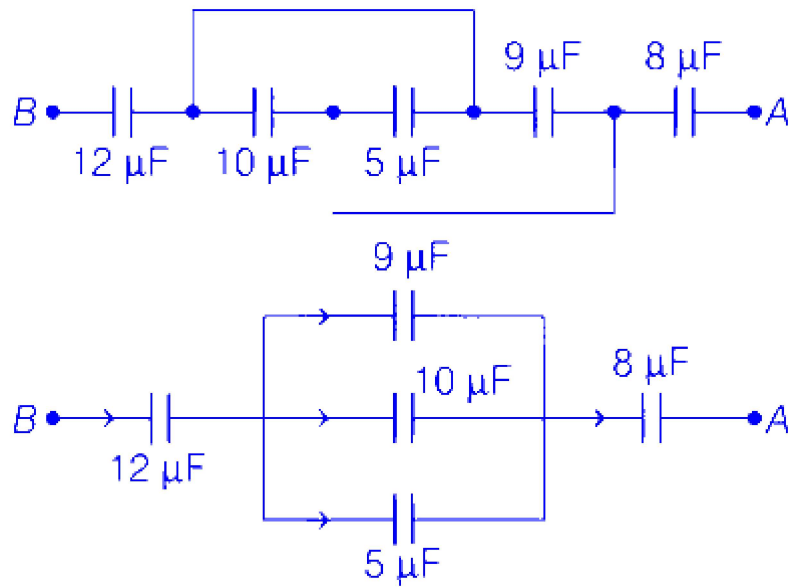
Options:

- A. $4 \mu\text{F}$ and $133 \mu\text{C}$
- B. $164 \mu\text{F}$ and $150 \mu\text{C}$
- C. $15 \mu\text{F}$ and $50 \mu\text{C}$
- D. $4 \mu\text{F}$ and $50 \mu\text{C}$

Answer: A

Solution:

According to given circuit diagram,



Voltage across A and B , $V_{AB} = 80 \text{ V}$

Let charge in $9 \mu\text{F}$, $10 \mu\text{F}$ and $5 \mu\text{F}$ be Q_1 , Q_2 and Q_3 .

Net charge, $Q = Q_{9\mu\text{F}} + Q_{10\mu\text{F}} + Q_{5\mu\text{F}}$

As we know that,

Parallel equivalent capacitance,

$$C'_{\text{eq}} = C_1 + C_2 + C_3$$

$$\therefore C'_{\text{eq}} = 9 + 10 + 5 = 24 \mu\text{F}$$

and series equivalent capacitance,

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_a} + \frac{1}{C_b} + \frac{1}{C_c}$$

$$\Rightarrow \frac{1}{C_{\text{eq}}} = \frac{1}{12} + \frac{1}{24} + \frac{1}{8} \Rightarrow \frac{1}{C_{\text{eq}}} = \frac{2+1+3}{24}$$

$$\Rightarrow C_{\text{eq}} = \frac{24}{6} \mu\text{F} = 4 \mu\text{F}$$

Since, $C = \frac{Q}{V}$

$$\therefore Q = CV$$

$$= 4 \times 10^{-6} \times 80 = 320 \mu\text{C}$$

Since, in series connection,

$$V = V_{12\mu\text{F}} + V_{10\mu\text{F}} + V_{8\mu\text{F}}$$

$$\Rightarrow 80 = \frac{320}{12} + V_{10\mu\text{F}} + \frac{320}{8}$$

$$\Rightarrow V_{10\mu\text{F}} = 13.33 \text{ V}$$

$$\therefore Q_{10\mu\text{F}} = V_{10\mu\text{F}} \times 10$$

$$\simeq 133 \mu\text{C}$$